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AIR ENTRY IN SALT MARSH SEDIMENTS

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The presence or absence of air in organic-rich peatland sediments is of profound importance in regulating sediment metabolism. It is not known how much, if any, air entry occurs in salt marsh sediments. We used pressure-volume techniques, dye tracers, and oxygen microelectrodes to determine the amount and distribution of air that enters sediments of Belle Isle Marsh in Boston, Massachusetts, U.S.A., under varying porewater pressures. The porewater pressure threshold for air entry averaged −18 cm (±5.4 cm, *n* = 5) of water at the marsh surface. Air entry increased approximately linearly with decreasing porewater pressure below this threshold and amounted to about 0.1% of sediment volume per cm H_2O of porewater pressure for cores averaging 10 cm in depth. As near-surface porewater pressure drops to −30 cm of water or lower during neap tides, air entry must periodically occur in Belle Isle Marsh. Field O_2 electrode measurements at porewater pressures averaging-26 cm of water also suggested about 1% of air-filled voids existed in the sediments, distributed in occasional "pockets" of millimeter dimensions. Hydraulic conductivity and dye tracer distributions could be explained by macropores in the effective diameter range of 60 to 200 μm. Many of these macropores seem to be associated with plant roots.

Knowledge of the extent and distribution of soil aeration is crucial to understanding the ecology and biogeochemistry of salt marshes. For example, the presence of oxygen profoundly affects sediment metabolism. King et al. (1985) indicated that, even though aerobic metabolism could be found only in the top 2 cm of a marsh sediment, it accounted for 45% of the carbon dioxide produced over a 15-cm-thick stratum. Aerobic respiration yields far more energy than does anaerobic metabolism via sulfate reduction or methanogenesis. The nature of the end products is important as well; reduced sulfur produced in anoxic sediments may be toxic, may form insoluble complexes, and represents an energy-rich substrate for chemosynthetic bacteria such as *Beggiatoa.* Oxygen availability in the soil also influences the vascular plant community; for example, Lindhurst (1980) found that the production of biomass of *Spartina alterniflora* in aerated systems was 6.3 times that of unaerated systems with all other growth conditions being equal. Not only are the bulk O_2 distribution and the timing of air entry important; O_2 distribution on a microscale may lead to a mosaic of oxidized and anoxic microsites whose existence and structure must be appreciated if the biogeochemistry of marshes is to be understood. Unfortunately, very few quantitative data exist on air entry in marshes of any type. We undertook the present study in a local salt marsh to determine (1) to what extent air entry occurs over the observed range of porewater pressure and (2) whether air entry was associated with particular structures (e.g., macropores) in the sediment. Because air and water movement and storage are intimately related, the necessary experimental procedures also yield ecologically significant information on related hydraulic characteristics such as hydraulic conductivity and elastic water storage.

BACKGROUND

Oxygen penetrates into organic marsh sediments by several mechanisms. Atmospheric oxygen may enter the sediment when the surface of the marsh is not inundated and the porewater pressure becomes less than atmospheric; however, porewater pressure must be sufficiently low to overcome the surface tension associated with soil pores at the surface (Childs 1969). The critical water pressure (P_{crit}), relative to atmospheric pressure, at which the surface tension may be overcome is:

$$P_{crit} = \frac{-2T}{R} \quad (1)$$

where T is the surface tension and R is the effective radius of the pore. Below this pressure,

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the porewater meniscus ruptures, the air-water interface retreats, and air is drawn in. The largest voids and channels will drain first, followed by smaller and smaller channels as porewater pressure continues to decrease. Because of the above relationship, any soil macropores that exist may be the first void spaces to drain as water is lost from a soil.

Soil macropores are water pathways, often of biotic origin, which are large in cross section compared to the effective pore diameters of adjoining soil. The existence of macropores has been well documented in terrestrial soils (see review by Beven and Germann 1982), and evidence that macropores can play an important role in water and solute transport in forest soils has been presented (Whipkey 1969; Weyman 1973). The dividing line between macro- and micropores is somewhat arbitrary. Bullock and Thomasson (1979) classified macropores as those larger than 60 μm in diameter, whereas Kanchanasut et al. (1978) use 200 μm for continuous vertical channels in silty loam. Bouma et al. (1977) defined macropores as relatively large soil pores—as water flows through them, a significant portion of soil volume is bypassed. A sediment dominated by macropore flow of gas or water presents a highly heterogeneous environment for plant roots and microbiota and may allow water or gas to penetrate along preferential flow channels much further into a sediment mass than would be expected in a homogeneous soil.

The physics of highly organic salt marsh soils have received little attention compared to those of agricultural soils, soils involved in geotechnical (e.g., foundation) designs, or forest soils. Models abstracted from such soils may or may not apply well to the structure of marsh peat. In visual examination of a plug of marsh soil, the textural role of plant parts (especially roots and rhizomes) stands out, suggesting that these structures may be important for air and water conduction.

The actual measurement of air entry into a marsh soil is confounded, to a degree not normally experienced in mineral soils, by the phenomenon of soil elasticity (elasticity as used here refers to the compressibility of the soil matrix in the vertical direction). Elasticity can be so high in a marsh soil that a certain amount of water loss is accommodated purely by a change in bulk sediment volume, with no desaturation occurring (Knott et al. 1987; Hemond and Fifield 1982). One relevant parameter for water storage by bulk soil expansion we call specific elastic storativity, *Se*. Change in water storage due to sediment elasticity is equal to the product: $Se \times$ depth of soil $\times$ change in porewater pressure. In measuring air entry by methods involving water drainage for marsh soils, it is essential also to measure peat elasticity to separate elastic water storage from water storage due to the draining and filling of soil pores.

In the present study we present data that pertain primarily to direct air entry. For completeness in an ecological context, we note that infiltrating water may also carry dissolved oxygen into the sediments. The importance of this process must be evaluated on the basis of the subsurface hydrology of the sediment (Rycroft et al. 1975; Hemond et al. 1984; Nuttle and Hemond 1988).[3] For this reason we include data on the hydraulic conductivity of Belle Isle Marsh peats (Table 1). Such conductivity data are essential to predictive modeling of marsh infiltration. Two mechanisms of sediment oxygenation not discussed here are molecular diffusion through the liquid phase and plant-facilitated gas transport (Dacey 1981).

METHODS

Study site

Experiments were conducted on the sediments of Belle Isle Marsh, located in East Boston, Massachusetts, west of Winthrop and northeast of Logan Airport. The average daily tidal range at Belle Isle Marsh is about 3 m. The peat at Belle Isle averages 2 m thick, the degree of decomposition increasing with depth. The organic content ranges from less than 5% to more than 14%. The root zone (0 to 60 cm) contains living and dead roots and rhizomes, the living roots and rhizomes penetrating to about 30 cm. From about 60 to 110 cm, the peat has a clayey appearance and individual plant remains are distinguishable only with difficulty. Underlying this partially humified zone is a third zone having a more clayey appearance, a finer texture, and few traces of plant remains. Below 2 m, a clay layer provides an apparently near-

[3] W. K. Nuttle, 1986, The flow of water in salt marsh peat, Ph.D. thesis, Dept. of Civil Engineering, Massachusetts Inst. of Technology.

impermeable boundary to water flow. Perhaps due to increased deposition of clay-sized mineral particles during dredging for the construction of Logan Airport, the texture of sediment between 6 and 11 cm deep is distinct from peat both above and below these depths throughout the study site.

The subsurface hydrological regime at this marsh is described by Nuttle[3] and by Nuttle and Hemond (1988). Briefly, the interior marsh experiences hydraulic head within a few centimeters of the marsh surface during spring tides. Head may drop to ~30 cm below the surface during nonflooding neap tides. Areas near the creeks experience low hydraulic heads on a diurnal basis because of gravity drainage at low tide. Water infiltration rates have been calculated for this marsh; however, the extent of air entry has not been heretofore investigated.

Field sampling procedure

Samples of marsh sediment were collected from a several-hectare area near the south end of Belle Isle marsh where the vegetation appears relatively homogeneous and undisturbed, being dominated by *Spartina alterniflora* and *Distichlis spicata.* Twenty-two sampling locations were chosen arbitrarily within this uniform-appearing area of marsh. Peat cores were taken in clear polycarbonate core liner (O.D. = 7.3 cm, I.D. = 6.7 cm). Shallow depths were cored by sharpening the core liner and pushing it into the peat. For deeper samples where the peat was more readily deformed and/or lost from the core liner, a piston coring apparatus of Knott et al. (1987) was employed. All cores were sealed with plastic end caps and tape upon recovery and were refrigerated upon return to the laboratory.

Hydraulic properties

The procedures described by Knott et al. (1987) were used to saturate the soil cores and to measure the hydraulic properties of saturated hydraulic conductivity (K_{sat}), compressibility index (a_v), specific elastic storativity (Se), coefficient of consolidation (α), bulk porosity (n), organic content, and specific gravity. Bulk porosity is operationally defined in this measurement on the basis of water loss upon drying; much of the "porosity" so defined is not necessarily available for bulk water flow. Specific elastic storativity, Se, is closely related to the compressibility index a_v by the relation $Se = (1 - n)\gamma_w a_v$, where γ_w is the weight density of porewater.

Air entry tests

To evaluate the relationships between the negative porewater pressures at the marsh surface and the occurrence of air entry, we used a laboratory test similar to that of Grable and Siemer (1968) (Fig. 1). The sample holder is a modification of the pressure cell of Debacker (1967) and Reginato and Van Bavel (1962). The porous stone has an air entry value of 0.5 bar, or 510 cm of water pressure. Below this pressure the stone, when wet, is porous to water but not air. The surface of the sample is open to the atmosphere but loosely covered with plastic wrap to reduce water loss by evapotranspiration. Deaerated water is used throughout the experiment.

At the start of each test, an effort is made to saturate the peat completely by immersing the sample in deaerated water for 24 hours, as described by Knott et al. (1987). The sample is then connected to the air entry apparatus. Initially adjusted to be level with the peat surface, the water level in the buret is subsequently lowered in steps. Lowered pore pressure causes water to be released from the peat. Several hours are allowed at each step for the system to reach equilibrium. Air entry is inferred by comparison of the water yield versus pore pressure curve with the corresponding compressibility curve for the sample. Generally, the two curves diverge at a well-defined inflection point, permitting straightforward interpretation of an air entry threshold. Five air entry tests were performed using surface cores taken in mixed stands of *Distichlis spicata* and *Spartina patens* at close proximity to a network of piezometers, so that the porewater pressures experienced by the cores in the field could be closely correlated to the range of pressures used in the air entry tests.

Dye tests

Rhodamine-WT (Smart and Laidlaw 1977; Nestler 1977) was used to obtain a qualitative representation of small-scale water movement in peat samples (Osmoti and Wild 1979). Clear polycarbonate tubing with a 7.6-cm inner diameter, with sharpened edges, was driven into the sediment to extract cores at least 20 cm long.

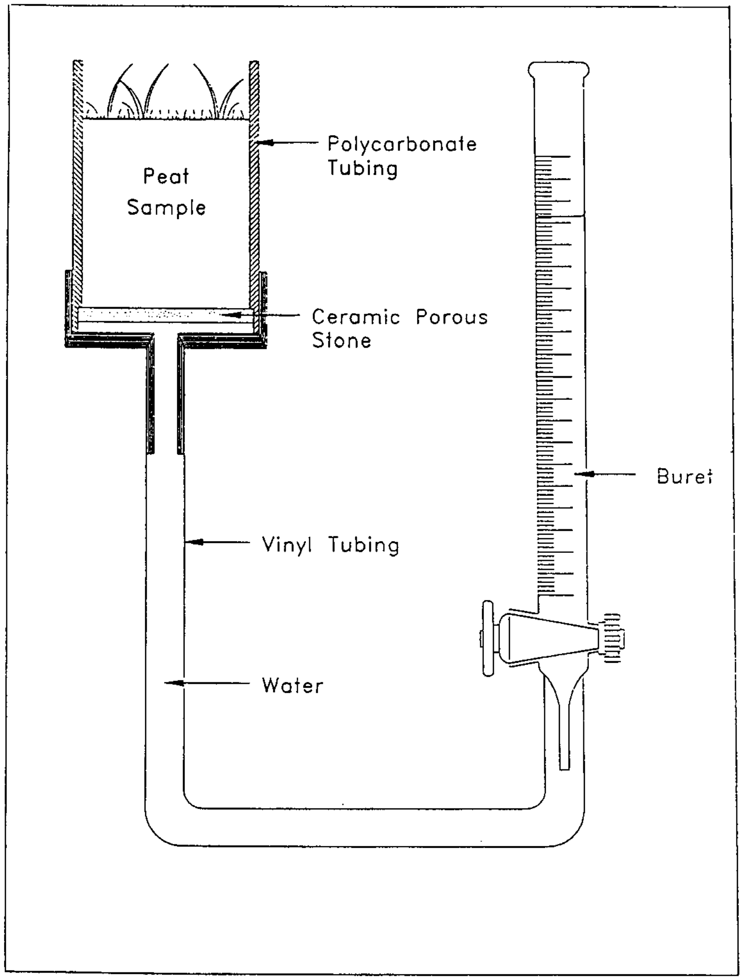

FIG. 1. Apparatus for air entry test. Elevation difference between water level in buret and peat surface corresponds, at equilibrium, to magnitude of pore pressure in sample.

Cores showing evidence of artifactual pores created by roots and rhizomes being dragged along the inner wall of the tubing were discarded. A ponding of rhodamine-WT at high concentration was added on the top of each core, allowing the dye to move into the core by gravity drainage. The peat samples were then extruded and sliced laterally at 2-cm intervals. The pathways traveled by the dyed water were clearly marked, and the dyed pores were usually correlated to specific visible structures in the peat matrix. Contact prints showing dye distribution at the faces of each slice were obtained by pressing the sediment slices against discs of white paper.

O_2 electrode

An oxygen electrode was made from approximately 1 cm of 0.5-mm-diameter platinum wire, crimped and soldered to a longer (about 20 cm) piece of #14 solid copper wire. The assembly was insulated on the sides by heat-shrink tubing and epoxy resin and was ground to a smooth, planar, circular surface at the tip. The resulting electrode assembly was sufficiently stiff and strong to be manually inserted into salt marsh sediments. An Ag/AgCl electrode (Skoog and West 1963) was used as an anode. A circuit was constructed to maintain electrode voltage at −0.67 volts while measuring electrode current (Forstner 1983). Subsequent analysis suggested that this potential could be sufficient to reduce H^+ if porewater pH dropped below 5.[4] However, the

[4] D. G. Chen, 1986, Air entry and flow through preferential pathways in salt marsh peat, M.S. thesis, Dept. of Civil Engineering, Massachusetts Inst. of Technology.

data presented here are used to demonstrate vertical patterns rather than to calculate absolute oxygen concentrations or diffusion rates. During field measurements the O_2 electrode was manually inserted into the sediment at a location no further than 5 cm from the anode, and the current was allowed to reach a steady value before the experiment proceeded to a subsequent depth. Pore pressures were simultaneously measured using a piezometer network.[3]

RESULTS

Hydraulic properties of Belle Isle sediment other than air entry are summarized in Table 1. The properties of the upper 60 cm of the sediment are of particular relevance because air entry is almost certainly restricted to this zone in the marsh. Mean porosity as measured by drying is high (0.84) and comparable to other marshes. Mean organic content (11%) is significantly lower than measured in other Massachusetts marshes (Knott et al. 1987) but is well predicted by specific gravity.[4] Saturated hydraulic conductivity is lower than measured in two other Massachusetts marshes by Knott et al. (1987), averaging 6×10^{-4} cm/sec for samples from the upper 60 cm. This is consistent with the lysimeter results of Nuttle[3] and with marsh-scale studies at this marsh (Nuttle and Hemond 1988). Hydraulic conductivity is lognormally distributed, as was found by Knott et al. (1987). The deeper peats have much lower hydraulic conductivities, the values consistent with the results of in situ slug tests conducted at a 1-m depth by Nichols.[5] Compressibility, a_v, averages 4.9×10^{-6} cm/sec$^2 \cdot$g, lognormally distributed, in the top 60 cm. Unlike hydraulic conductivity, compressibility decreased only slightly with increasing depth. The corresponding mean value for specific elastic storativity, Se, was $6.9 \times 10^{-4} \pm 3.3 \times 10^{-4}$ cm^{-1} ($n = 22$).

Air entry tests

The pore pressures chosen for air entry tests started at 0 and ranged to as low as -40 cm of water in steps of 2 or 5 cm of water. All peat samples were 6 to 12 cm long. Fig. 2 shows a typical result. For relatively high pore pressures (0 to about -10 cm), the slope, S_{S1}, of the water content versus pore pressure curve averaged $8 \times 10^{-4} \pm 1.5 \times 10^{-4}$ cm^{-1} ($n = 5$), very close to the average specific elastic storativity. This correspondence suggests that little desaturation occurs in this range of porewater pressure.

The air entry value, P_{crit}, is the porewater pressure at which air begins to penetrate the peat matrix. We interpret the inflection points of the drainage versus porewater pressure curves as a measure of this pressure. P_{crit} ranged from -13 to -27 cm of water, averaging -18.2 cm of water and having an arithmetic standard deviation of 5.4 cm. The mean value of slope of the water content versus pore pressure curves for pore pressures lower than P_{crit}, a value we call S_{S2}, is 2.0×10^{-3} cm^{-1}, much greater than can be accounted for by sediment compression. (The compressibility curve itself is essentially linear over the range of interest.) The inflection point is visually well defined for each test. The difference between the two slopes, S_{S1} and S_{S2}, yields a measure of the volume of air entry as a function of porewater pressure when porewater pressure becomes less than P_{crit}.

Dye tests

Sediment cores infused with dye generally exhibited (1) an upper region in which all material was heavily dyed and no individual flowpaths could be distinguished; (2) a region in which discrete structures were dyed, suggesting preferred flowpaths; and (3) a region to which the dye had not penetrated. Fig. 3 presents tracings of a sequence of dye prints from a single core. Inferred preferential flowpaths were associated with roots and rhizomes, the structures having estimated typical diameters of 300 and 1000 μm, respectively. In region (1) it was not possible to tell whether the uniform dyeing was due to homogeneity of the medium or to diffusive lateral mixing of dye from flow pathways. Individual plant roots were most readily distinguishable below 6 cm.

O_2 electrode data

Electrode current indicated the presence of dissolved oxygen at the marsh surface. However, electrode current rapidly decreased with increasing depth, and current was undetectable below depths that ranged from 0.5 to 6 mm, depending on site (Fig. 4). Within this thin near-surface region, little correlation between porewater

[5] E. M. Nichols, 1985, Determination of the hydrologic parameters of salt marsh peat using in situ well tests, M.S. thesis, Dept. of Civil Engineering, Massachusetts Inst. of Technology.

TABLE 1

Summary of several bulk hydraulic properties of Belle Isle peat[a]

	Root zone (0–60 cm)	Partially humified peat (60–100 cm)	Clayey peat (110–200 cm)
log hydraulic conductivity (log K) (cm/s)	-3.2 ± 0.37	-4.0 ± 0.65	-5.2 ± 0.68
Compressibility (a_v) (cm-s^2/g)	$4.9 \times 10^{-6} \pm 3.1 \times 10^{-6}$	$3.6 \times 10^{-6} \pm 2.8 \times 10^{-6}$	$3.2 \times 10^{-6} \pm 2.6 \times 10^{-6}$
Specific elastic storage (Se) (cm^{-1})	$6.9 \times 10^{-4} \pm 3.3 \times 10^{-4}$	$6.5 \times 10^{-4} \pm 4.0 \times 10^{-4}$	$7.2 \times 10^{-4} \pm 5.0 \times 19^{-4}$
Coefficient of consolidation (α) (cm^2/s)	1.3 ± 1.1	0.25 ± 0.48	0.02 ± 0.02
Porosity (n)	0.84 ± 0.05	0.76 ± 0.08	0.76 ± 0.04
Organic content (%)	11.0 ± 3.1	5.8 ± 1.9	5.8 ± 0.70
Specific gravity	2.4 ± 0.18	2.6 ± 0.11	2.6 ± 0.13

[a] Values presented are means ± standard deviation, for a population of 22 cores, taken at arbitrary sampling locations throughout a vegetationally homogeneous area of the marsh.

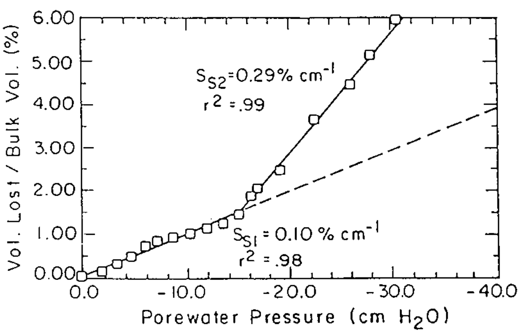

FIG. 2. Plot of water released versus pore pressure for a typical sample of surface peat from Belle Isle marsh. In the range of 0 to -18 cm H_2O the water released can be accounted for by the compressibility of the sediment. Below 18 cm H_2O, water is released at a significantly greater rate because of the additional process of air entry. Uncertainty in volume of water release is of the order of ±0.05 ml, which results in vertical error bars on each datum equal to 0.014%, a value smaller than symbol height.

pressure and the depth at which oxygen became unmeasurable was evident, suggesting that the peat surface is rather heterogeneous. Below the thin surface layer, most of the sediment seems to be essentially anaerobic when the water table is within 30 cm of the surface. At four sites, however, there were short depth intervals in which electrode current rose substantially above background, suggesting the presence of small, oxygenated "pockets" (Fig. 5). These short depth intervals of high electrode current occurred several centimeters below the sediment surface.

DISCUSSION

The slopes, S_{S1} and S_{S2}, of the air entry test represent the water storage properties of marsh sediment in particular pressure ranges. The sharp break between slope S_{S1} and slope S_{S2} is suggestive of a transition from purely elastic water release (which must occur at all pressures and is essentially linear over the pressure range studied) to a region where combined elastic and desaturation (air entry) mechanisms operate. Nuttle proposed that the total specific storativity of marsh sediment, S_s, contains the following components[3]

$$S_s = S_s^B + S_s^R + S_s^G + S_s^{AE} \qquad (2)$$

where B, R, G, and AE refer to storativity associated with bulk matrix compressibility, root/rhizome compressibility, isolated gas bubble compressibility, and air entry, respectively. The sum of the first three terms is equal to the specific elastic storativity, Se, described earlier. From observations of marsh surface movement, Nuttle estimated that Se is about 7.0×10^{-4} cm^{-1} at Belle Isle.[3] This is in good agreement with our mean value of 6.9×10^{-4} cm^{-1}. It is not distinguishable from the mean slope S_{S1} observed in the air entry tests (8.0×10^{-4} cm^{-1} ± 1.5×10^{-4} cm^{-1}). Our interpretation is that S_{S1} is mainly due to the sum of S_s^R, S_s^B, and S_s^G and represents predominantly compressive storage; any air entry in this pressure range would be much less than the water loss due to peat compression.

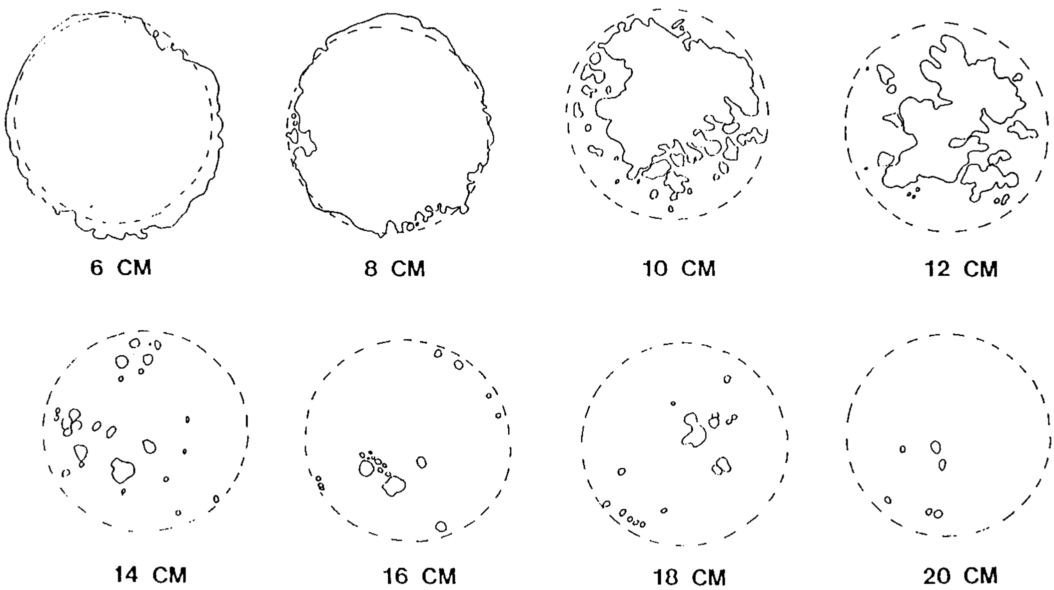

FIG. 3. Tracings of dye obtained along the length of a sediment core into which dye was infused at the surface. Encircled areas represent areas of dye coloration. (Dye appears to lie outside the cross section at 6- and 8-cm depths because the dye spread on the paper. The spreading was less problematic at the deeper cross sections where dye concentrations were lower.) For cross sections below 12 cm, dye occurred in distinct spots associated, usually, with plant roots.

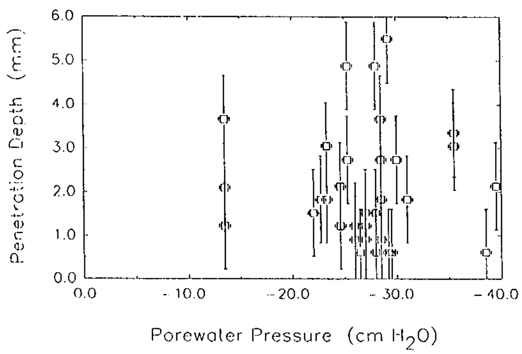

FIG. 4. Oxygen penetration depth, measured by polarographic oxygen electrode, as a function of porewater pressure at 33 marsh sites. Vertical and horizontal error bars correspond to the typical ±1 mm vertical resolution of electrode placement and to piezometer readability, respectively. Despite sediment heterogeneity, anoxic sediment was encountered at a depth of 6 mm or less in all cases. However, no correlation could be found between O_2 penetration depth and porewater pressure in these data; the correlation coefficient is only 0.03.

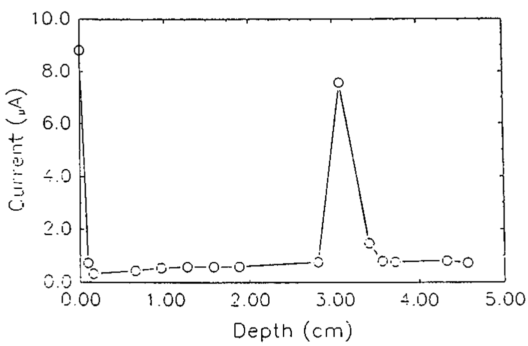

FIG. 5. Example of an apparent oxygen "pocket" in marsh peat, measured in situ with a polarographic oxygen electrode. Near-surface porewater pressure at the time and place of measurement was −28 cm H_2O. Resolution of current and depth measurements is of similar magnitude to symbol size.

P_{crit} is the porewater pressure corresponding to the beginning of air entry into the sediment. Such air occupies voids initially filled with water. Thus, for pressures below P_{crit}, an increased amount of water release occurs in addition to that released due to compression of matrix, roots, and rhizomes and expansion of isolated gas bubbles. The specific storage due to air entry, S_s^{AE}, is taken as the difference between S_{S1} and S_{S2} for porewater pressures lower than P_{crit}.

$$S_s^{AE} = S_{S2} - S_{S1} \quad (3)$$

The average value of S_s^{AE} for the samples tested is 1.2×10^{-3} cm^{-1} $\pm$ 0.7×10^{-3} cm^{-1}, $n = 5$. The contribution from air entry is very significant beyond P_{crit}, comprising over 60% of the storage coefficient for the peat samples studied.

In interpreting these results in an ecological context, we should note that, for similar sediments, the importance of air entry is greatest in shallower sediment deposits. For any given change in porewater pressure, the amount of water lost or gained per unit area of marsh increases in direct proportion to peat depth, whereas saturation/desaturation and associated air entry does not increase. Sediment below the water table does not participate in changes of saturation, regardless of macropore dimensions.

Capillary tube model

In an effort to explore the measured air entry characteristics of Belle Isle peat in terms of their consequences for peat structure, we invoked a simple capillary tube model. There are several assumptions underlying the model. First, the air entry pressure for each pore is inversely proportional to the diameter, D, of the pore. Second, pores are assumed to be vertical, continuous, and independent with cross sectional areas varying from pore to pore but constant along each pore (like a bundle of drinking straws). Third, whenever the surface porewater pressure of a sample is more negative than the critical pressure of a pore, the entire pore is assumed to drain. It is assumed that there is no exchange between the capillary tubes and the peat matrix. Finally, the surface tension of porewater is taken as that of pure water.

For each increment of porewater pressure change, the volume of water ΔV_{ae_i} drained from the model due to desaturation is given by

$$\Delta V_{ae_i} = \frac{\Delta N_i \pi D_i^2}{4} L \tag{4}$$

where ΔN is the incremental number of pores drained, L is the length of the sample, and D_i is the diameter of the pores drained. Combining with Eq. (1) gives

$$\Delta N_i = \frac{\Delta V_{ae_i} P_i^2}{4\pi T^2 L} \tag{5}$$

Mean values from the air entry tests are used to generate the pore size distribution of Fig. 6. In this model there are many more pores in the 70- to 160-μm range than there are pores larger than 160 μm.

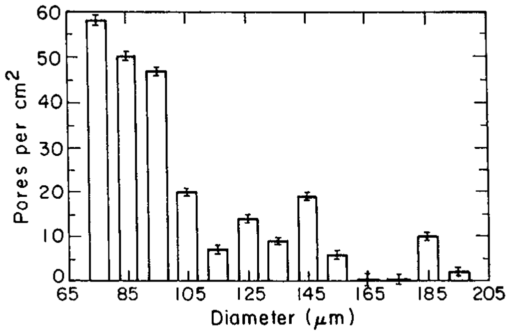

FIG. 6. Calculated pore-size distribution corresponding to mean peat air entry characteristics and the assumptions of a simple "bundle of drinking straws" model. Dye test, conductivity, and oxygen electrode data collectively suggest a different pore structure, consisting of larger voids interconnected by a network having a similar effective diameter distribution but lower areal density of pore spaces. Error bars represent uncertainty propagated from the uncertainty in water yield for each increment of pore pressure imposed on the sample.

Hydraulic conductivity of the capillary tube model

One test of the capillary tube model is its ability to predict saturated hydraulic conductivity. The hydraulic conductivity of a bundle of tubes can be calculated assuming that Poiseuille flow at velocity v occurs in each tube according to

$$v = \frac{\pi p r^4}{8 l \eta} \tag{6}$$

where p is pressure difference across the tubes, r is the tube radius, l is tube length, and η is the viscosity of water. The resulting expression for hydraulic conductivity, given water at room temperature, is

$$K = \sum_i 3.8 \times 10^4 \, r_i^4 N_i \text{ (cm/sec)} \tag{7}$$

where r is in cm and N_i in cm^{-1}. The hydraulic conductivity calculated using the pore diameter distribution of Fig. 6 is 10^{-2} cm/sec, more than an order of magnitude greater than measured values. This demonstrates that the capillary tube model is an oversimplification, perhaps in regard to uniformity, continuity, and/or orientation of pores. Orientation is unlikely to be the major shortcoming of this model, however, because several in situ dye tests, conducted as described earlier but with the core tubes imbedded in the marsh, revealed little lateral spreading of dye in sediments below the constraining influence of the core tube.

Density of macropores from dye tests

The simple capillary tube model predicts that macropores having effective diameters of 70 μm or greater occur in Belle Isle peat at densities in excess of 200 per cm^2. Observations of dyed peat sections, however, suggest a lower density of discrete flow pathways, at most a few tens per cm^2. Dye test results thus agree with the hydraulic conductivity calculations in suggesting that the simple capillary tube model overestimates the density of macropores in the marsh sediment.

O_2 microelectrode data

The 33 oxygen electrode profiles obtained in this study suggest the presence of occasional oxygen "pockets" in marsh sediment. The vertical extent of the pockets typically seems to be 1 mm or more. Treating the microelectrode as a point (assuming it to have an area much smaller than the voids it encounters) and following Fluhler et al. (1976) (who consider the statistics of anoxic volumes in otherwise aerobic soil; here we reverse the oxic-anoxic pattern), the most probable volume of oxic pockets in the sediments is about 2%. At a pore pressure of −26cm, typical of the conditions under which many of the in situ O_2 electrode measurements were made, average sediment is predicted on the basis of air entry tests to contain about 1% air. These values cannot be argued to be significantly different, given the few pockets observed and the heterogeneity of the sediment.

The observed air entry and hydraulic conductivity characteristics of Belle Isle surface sediments can be reconciled by a conceptual macropore model in which most of the void space occurs as partially isolated pockets with characteristic effective dimensions of the order of millimeters but playing a minimal role in hydraulic conductivity. These voids, corresponding to the O_2 pockets inferred from O_2 electrode data, are connected into a network of flow tubes with a relative diameter distribution similar to that of Fig. 6 but an areal density an order of magnitude lower. This is consistent with the low densities of macropores inferred from the dye tests. A further test is comparison of predicted versus measured *unsaturated* hydraulic conductivity. The "bundle of straws" model, revised only with respect to flow tube areal density, predicts a typical conductivity of 2×10^{-4} cm/sec at a pore pressure of −22 cm H_2O, whereas Nuttle (1990) calculated a value of 5×10^{-4} cm/sec^{-1} from lysimeter data for Belle Isle sediment at this pressure. The presence of soil gas occurring in pockets was also suggested by the observations of Chapman (1974) on the basis of data from a manometer that he indicated was attached to one such pocket.

CONCLUSIONS

In its hydrological properties, Belle Isle marsh sediment is of lower hydraulic conductivity and compressibility than sediments in two other Massachusetts marshes studied by Knott et al. (1987), perhaps due to a higher content of fine mineral material (clays). The air entry threshold averages about −18 cm of water, whereas porewater pressures drop well below this value at the surface during neap tides (see Nuttle and Hemond (1988) for a discussion of the hydrology of this marsh). Both air and water flow seem to occur primarily along preferential pathways, typically associated with plant roots. Substantial nonuniformities in these pathways must exist because neither hydraulic conductivity nor the observed density of the dye-marked flow pathways matches predictions of a uniform tube model generated from air entry data. Inferred volumes of soil gas range up to about 2%, similar to that estimated by Nuttle on the basis of compressibility.[3] Because of this high degree of subsurface heterogeneity in marsh sediments, biota (microbes and plant roots) seem likely to be spatially organized in relation to associated gradients of oxygen. Preferential association of roots with sometimes oxic macropores could create a self-perpetuating macropore structure in marsh soils. We further suggest that, because of preferential flow pathways, efforts to sample pore water may yield samples biased by an overrepresentation of sometimes oxic microenvironments associated with macropores. Further work to elaborate on marsh sediment macropore structure is needed, and application of the extensive body of macropore concepts developed for terrestrial ecosystems may prove fruitful in marsh ecosystems.

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